

OnStage: Design and Development of a Performance and Practice Simulation System

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ABSTRACT

Traditional dance practice processes use mirrors or videos to observe movements and facial expressions, have limitations due to their reliance on a single viewpoint. Meanwhile, the transition from dance training rooms to stage performances involves significant environmental changes, including limited rehearsal times and limited opportunities for costume adjustments. In this paper, we present a dance performance and practice simulation system that allows dancers to observe dance movements from multiple perspectives, providing them with an advance preview of their performance in a formal setting.

Index Terms: Human-centered computing—Interactive systems and tools—;—Computing methodologies—Virtual reality—

1 INTRODUCTION

Stage performance is a highly practical art form, and performers need to enhance their skills through daily training to effectively convey emotions to the audience. However, due to practical reasons such as budget constraints on costume and stage rentals, performers often receive such information only a few days before the formal performance. If complex costumes are involved, as in Chinese classical dancing, stage accidents, such as accessories falling and performers stepping on costumes, frequently occur during formal performances. Moreover, there are significant environmental changes from rehearsal rooms to stage performances. Lighting and stage size can easily influence dancers, hindering them from performing at their highest level.

In recent years, with the development of human-computer interaction and 3D simulation techniques, some intelligent dance assistance systems have been applied in dance creation [2] and practice [3], providing effective computational assistance for creators and performers. One of the most commonly used techniques in stage performance practice is motion capture [6], involving the recording of dance data to assist dancers and researchers in training and choreography [5]. Many studies combine virtual reality technology with motion capture techniques to support dance practice and education [1]. Digital simulation has opened up new possibilities for drama, dance, and other artistic works. It can aid in generating virtual performance scenes and stages [4]. Based on these perspectives, we emphasize the need for more visual and aesthetic applications of motion capture results in the field of stage performance art.

This paper takes dance performance as a case study and addresses two issues using simulation techniques. On one hand, we aim to assist performers in their expression through motion capture, facial expression capture. On the other hand, we strive to provide performers with more on-stage performance information in advance, allowing dancers to simulate the formal stage performance environment using a virtual engine. We conducted an initial survey with

professional dancers to understand their challenges in these two aspects. Subsequently, we designed the *OnStage* system to help dancers simulate their performance on the stage (as shown in Fig. 1), allowing performers to integrate into the formal stage environment, observe their movements and expressions from multiple angles, and help them simulate their performances on stage.



Figure 1: Performance simulation using *OnStage* system.

2 USER STUDY

To address our research questions and gain a better understanding of the target users for the dance virtual simulation and practice system, we conducted a survey with dancers and students from professional dancing schools. We collected a total of 58 responses from the online questionnaire. Additionally, we summarized the questions from face-to-face interviews with dancers in a university dance team, aiming to understand the dancers' requirements for stage performance simulation and their attitude toward the simulation system. We collected basic information from the participants to confirm that they have extensive performance experience.

2.1 Assisting Technologies for Dance Performance

All participants expressed concerns about the amplitude of their dance movements and the coordination between facial expressions and dance movements on stage due to the larger performance space, worrying that their dance movements might be too large or too small.

2.2 Stage performance simulation

All participants believed that obtaining information about costumes and stage effects in advance could prevent distractions on stage, providing them with sufficient mental preparation. Also some interviewees think it is very useful to simulate stage effect before the performance but sometimes the effect is not controllable.

2.3 Attitude toward simulation system

98.28% of the participants expressed a willingness to try the virtual dance simulation and practice system (Neutral or above), which has given us great confidence.

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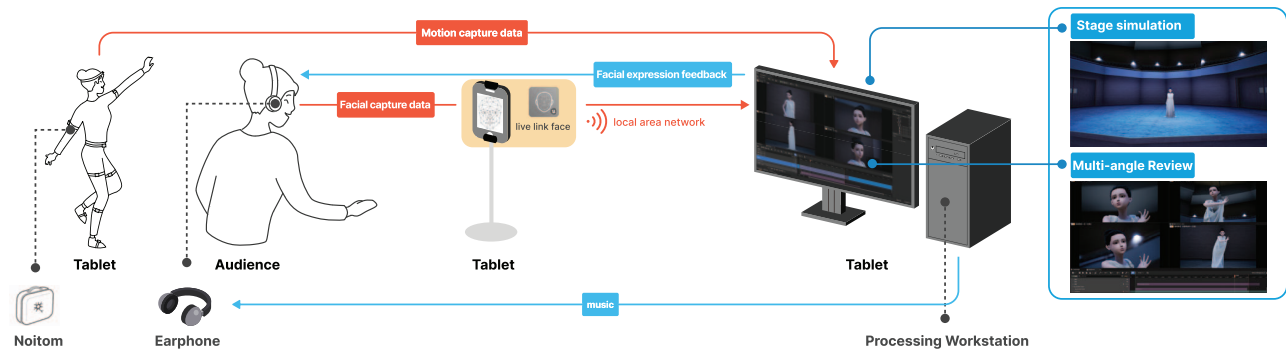


Figure 2: System Overview of *OnStage*. The system contains three functions: Motion and Facial Expression Capture, On-stage Simulation and Multi-angle Review.

3 SYSTEM DESCRIPTION

3.1 System Design

Based on user research, we designed the *OnStage* system. The system aims to facilitate dancers in adapting to the stage in advance, understanding the performance effects, and assisting them in adjusting the amplitude of movements and facial expressions during daily training. The system comprises three functions as shown in Fig. 2.

The **Motion and Facial Expression Capture** function records dancers' movements and facial expressions during practice and stores them for further 3D simulations. The captured results are simultaneously mapped onto virtual characters, thereby simulating the dancer's performance. This allows dancers to thoroughly examine their dance movements, ensuring that facial expressions accurately convey the desired emotions.

The **On-stage Simulation** function reproduces costumes and stage environments, enabling dancers to experience formal performances in advance. This includes simulating elements such as costumes, lighting, and stage size. Dancers can experience the ambience of a formal performance in advance. Dancers can provide feedback to costume and stage designers based on the simulated effects, allowing for better adjustments and optimizations of performance details.

The **Multi-angle Review** function allows dancers to review their recorded practice from different viewpoints. For example, dancers can use a frontal facial area window to review their facial expression details. They can also simulate their performance from the audience perspective.

3.2 Implementation

The *OnStage* performance and practice simulation system was built with a motion capture kit, a smart tablet and a processing workstation. The system was developed and operated on a workstation with an Intel i7 CPU at 2.4GHz, an NVIDIA GeForce 1080GTX GPU and 8GB RAM. For motion capture, we chose to use Noitom Perception Neuron 2 kit, which was wirelessly connected to the processing workstation running Axis Neuron software. Dancers' movements were stored in FBX format for subsequent character animation control. We used an iPad running the ARKit Live Link Face application for facial expression capture, and the signals were transmitted directly to the workstation through a local area network.

The character and stage was created with Blender software while costumes were created with Style3D software. Lighting, costumes, and motion simulation are created in the Unreal Engine 5.0. Utilizing the lighting system in Unreal 5.0, we designed the stage lighting, replicated the stage size and arrangement, and rendered the entire stage environment.

3.3 Procedure

Users are first guided on how to use the *OnStage* system, including the use of the system interface and how to record and review their dance movements and facial expressions. Subsequently, they need to wear motion capture and facial capture devices, perform designated dance segments in the *OnStage* system, and record their performances. After the performance, they can choose viewpoints from different angles to observe their own performance.

4 CONCLUSION

We design and implement a practice and perform simulation system, encompassing facial expression and motion capture, costumes, and virtual simulation of the stage environment. The system aims to assist performers in adapting to the stage environment in advance, adjusting the amplitude of movements and facial expressions, minimizing issues arising from environmental changes, and thereby preventing stage accidents.

In future work, we plan to enhance the sensitivity of facial expression capture, integrate dance feedback using facial and motion recognition technologies, and introduce adjustable parameters for users to fine-tune motion amplitude and dance environment. Additionally, we hope to polish our system with more light-weighted motion capture and face capture hardware. Furthermore, we intend to evaluate the system's usability through user experiments.

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